

EgoScore — A Curation Engine for EgoVerse

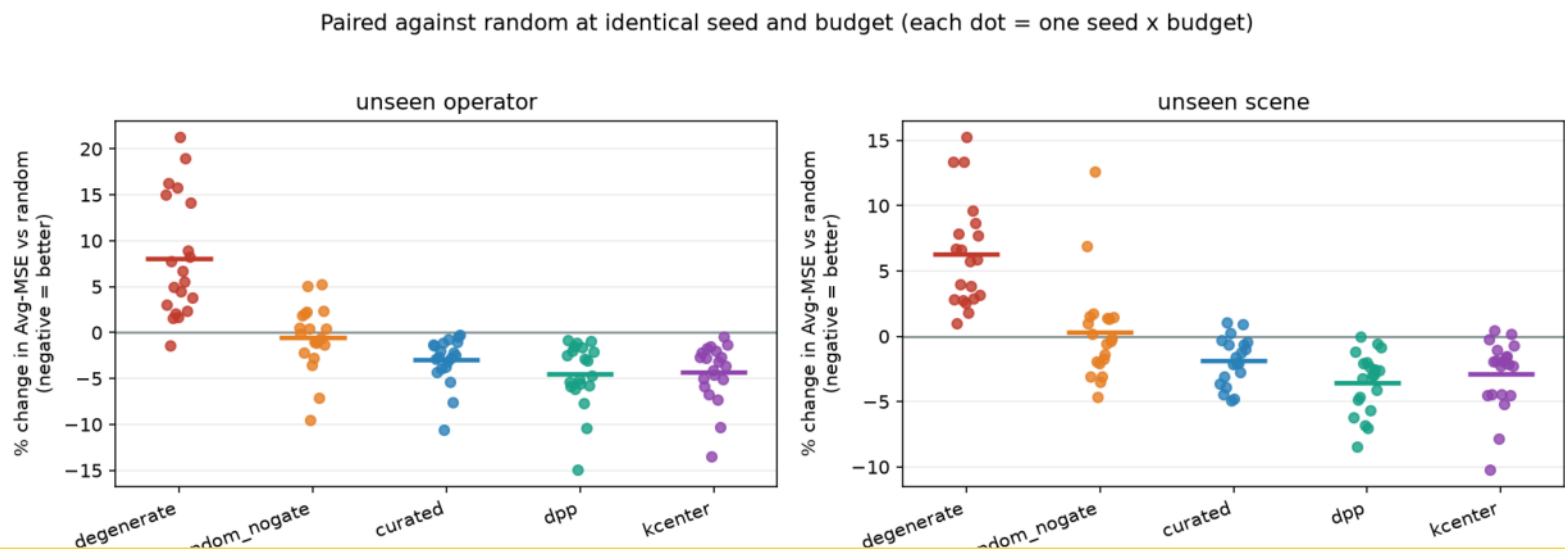
Track 1 · Which episodes are worth training on? · github.com/sahilmahendrakar/egoscore

Claim tested: at a fixed budget K , a quality-gated, coverage-maximizing subset trains a better policy than K random episodes.

Result

10 seeds × 2 budgets × 2 held-out axes = 40 paired tests

SELECTOR	WINS	AVG-MSE	p
dpp (log-det diversity)	40/40	-4.03%	2e-12
kcenter	38/40	-3.61%	1e-11
curated (facility location)	37/40	-2.44%	1e-09
no quality gate	22/40	-0.11%	n.s.
degenerate (control)	1/40	+7.17%	9e-12



Not “throw away 75% of your data for free” — the full gated pool still wins.
At ¼ budget, diversity selection closes 56% (unseen operator) and 50% (unseen scene) of the gap between a random quarter and using everything.

Why the control matters

A few percent on a proxy metric is easy to disbelieve. So we added a condition that should lose: the same budget concentrated into ~10 operator×scene groups instead of ~60.

It loses 1/40, +7.2% — large, unambiguous, correctly signed. The harness detects the diversity effect the EgoVerse paper reported, so the smaller selection margin is a measurement, not noise.

Selection matters here. Filtering does not: the quality gate is n.s. on data this clean.

Three things we found in the data

Zarr arrays are zero-padded to a chunk boundary. Read them raw and the trailing frames look exactly like a frozen tracker — the gate would have discarded the entire dataset. True length is total_frames in attrs.

Images cost 84.6 GB for this slice; poses cost 2.3 GB. Every signal here is pose-only, deliberately: curation must cost less than the training it saves.

An “episode” is not a common unit: median 93 s in rl2 vs 6.6 s in mecka. Episode-count budgets do not transfer across labs.

Limitations

Avg-MSE is a proxy, not robot success — the EgoVerse authors say so themselves, and we adopt it on those terms.

The proxy policy is proprioceptive, not visual. One task, one lab — no claim of transfer.

Facility location was our a priori pick and it lost to DPP and k-center. Reported, not quietly reordered.

Method: rl2 fold_clothes — 572 episodes, 20 operators × 16 scenes, the only EgoVerse slice with a populated operator×scene grid (microagi has no operator/scene metadata; mecka has 2 scenes).
Proxy: offline Avg-MSE (the EgoVerse authors' own metric) from a ridge action-chunk policy — 10-frame proprio history → 30-step future EE pose, relative to current pose.